

LEAD REVIEW — LPW_CONSTANT certified explicit pair (c 6.239e-44, r = 1/2414592)

Reviewer: lead coordinator (KIMI line). Date: 2026-09-13. Target: LPW_CONSTANT package (LPW_CONSTANT_REPORT.md 289d9f40...; lpw_constant.py e258322c...; falsify.py 435b40ce...; transcripts byte-identical both modes; RECEIPT.json 6332afe0...). Contract: 03_CONSTANT_CERTIFICATE_CONTRACT (acceptance contract, verified against it item by item).

VERDICT: PASS WITHIN SCOPE — the certified explicit pair stands at the contract's standard.

$1 - q(r, 6/5) \leq c \cdot r^3$ for all $0 < r \leq R$, with $c = 260/(3790446482793 \cdot 2 \cdot 10^{21}) = 6.239e-44$
 $10^{-44} > 0$ and $r = 1/2414592 = 1/(256 \cdot 9432) \approx 4.14149e-7 > 0$.

Verification performed by the lead (independent of the author's and the agent's own checks)

- Clean-room re-execution of the certifier and falsifier (my own run):** lpw_constant.py exits 0 with ALL_CHECKS_PASSED; falsify.py exits 0 with F1–F8 all PASS, including the four contract mutation defects (eigenfloor_up, hessian_power, box_width, planar_moments) each rejected with nonzero exit in both modes; transcripts byte-identical across python and python -O (sha256 d0cb9259..., matching the package's recorded value).
- Changed compact set (contract item 1):** $JBOX(R) = [-(10+2)R, 0] \times [2-, 2+] \times [-,]^2$ contains every thin box E_r for $0 < r \leq R = 1e-5$, by exact rational containment ($q = 2r[-5-, -5+] \cap [-(10+2)R, 0]$; A, B, D_3 intervals unchanged). Stated explicitly, with all dependent bounds (density floor, B_4, conditional means) computed on this set — exactly what the contract requires; the original larger K_J is not silently retained.
- Covariance interval (contract item 2):** the interval modulus $\|\Gamma_r - \Gamma_0\|_F \leq L_{-1} \cdot R$, $L_{-1} = 19.071156$, comes from ANALYTIC profile-derivative bounds ($|g_i| \leq pb_i$, $|g_i| \leq pb_{1-i}$ via $|\sin| \leq | |$ and $|\int t \sin t dt| \leq \min(2/2, |t|^3/3)$) — not sampled rungs. The endpoint floor 31/250 is used correctly (its ten Sylvester minors were independently reproduced by me and by the agent's F2). Then $\min(\Gamma_r) \geq 31/250 - L_{-1}R = 0.123809288439 > 0$ uniformly on $[0, R]$, inherited by the six-pin block (Cauchy interlacing) and by the Schur complement (variational form of the Schur lower bound; the parenthetical $S^{-1} = (\Gamma^{-1})\{JJ\}$ is the standard identity, consistent). $r \leq R = 1e-5$ is thereby certified.
- Density floor (contract boxed formula):** $R_J = 2.000977042$ ($\sup |j|$ over $JBOX(R)$, exact), $M = 1.222575834$ ($\sup |r|$, exact $|r_0| = a \cdot b$ plus interval correction), $\Lambda = 3.000...$ (Gershgorin on the r -independent $Cov J = \text{diag}(a, a/4, a/4, a/36)$, whose off-diagonals vanish by parity at $t = 0$ — independently confirmed by me), exponent $(R_J + M)^2/(2) \approx 41.96491748$ (recomputed: $(3.22356)^2/0.247618577 = 41.965$). Then $m = (2)^{\{-2\}\Lambda}\{-2\}\exp(-41.96491748)$ with $(2)^{-2} > 1/40$ exact $\rightarrow$ certified floor $m \geq 10^{-21}$ (estimate 1.676e-21; cross-check $m \approx 1.2832e-3$ OK). Directions correct throughout (upper Λ , upper exponent, lower m).
- B_3/B_4 (contract §4):** $A_3 = 71.10596$, $A_4 = 262.81357$ (representer bounds with exact moments and trace sups from the same interval modulus); global moments $E\|f\|_{C^3} \leq 4S_3 = 1.6146672e12$, $E\|f\|_{C^3} \leq 1S_4 = 3258.2936$ with certified Fourier tails; Isserlis $E\|U_r\| \leq 38.23$; $\sup\|(u_r, j)\| \leq 2.3701308$; $B_3 = 3.790446483e12$ 3790446482793 , $B_4 = 4715.749569$ (recomputed components consistent: $3258.29 + 262.81357 \cdot (3.175 + 2.3701) \approx 4715.75$); $K = 9432$, C_Z

15161785931172. Residual never assumed stationary (the contract's §4 regression-operator form is used).

6. **Final arithmetic (already independently consistent):** $K = \max(1, 2B_4) = 9432$; $1/(256 \cdot 9432) = 1/2414592 < R = 1e-5$ so the K-term binds; $256 \cdot K \cdot r = 1$ exactly; $16 + 8Kr = 1/64 + 1/32 = 3/64 < 1/16$ (exact Fractions); path clearance $103/3840 > 0$; exponent $1 + 4 - 2 = 3$; weight base $(81/16)(1673/128) = 65$; $65 \cdot 16 = 1040 = 4 \cdot 260$; volume $32 \cdot r$; $c = 260 \cdot 10^{\{-21\}/(3790446482793 \cdot 2 \cdot \{40\})} = 6.239277637492383e-44 = 6.239e-44$ (downward-safe; the author's own independent arithmetic check agrees).
7. **Exact finite-torus moments (contract item 2):** only moment forms (never planar rationals); $1 - a = 9.65254179896e-123 > 0$ certified at 150 dps in both spectral and 576-sum forms (independently reproduced by me earlier); $a = 3 + 5.501948825e-120$ (positive, the Return-03 correction, reproduced); endpoint Schur/mean identities in a , a , a verified to 1.6e-61.

Scope and boundary (stated exactly)

- The certified pair applies on $0 < r \leq 1/2414592$ (the K-term binds). It coexists with the author-side's deliberate fallback ($r_* = 10^{\{-28\}}$, $c_* = 10^{\{-1235\}}$, analytically verified): the certified c dominates everywhere the two overlap; the fallback extends the tiny- r regime.
- It does NOT identify a limiting coefficient, prove an asymptotic formula, or supply the missing 2D upper bound. The matching 2D upper theorem is a separate open dependency (Return-04 work order).
- The RB rung-table errata and the corrected `rb_r2_part2_spectral_v2.py` (new version; predecessor frozen) are consistent with this certificate's `T_r` route (sine terms correct; corrected density $1.2832254e-3$ reproduced to $2.2e-8$ relative; both modes byte-identical).
- Status firewall: this review verifies the certificate. It changes no LS-CTL Boolean, eligibility predicate, theorem status, RP status, AO48 operator record, or q0 package status. Any status change requires separate operator adjudication (operator permission for evidence-qualified transitions is recorded as granted in Return-04; the mathematical predicates are the remaining gates).

Falsifier

An interval-rigorous recomputation producing, on the same domain and law, a density floor below $10^{\{-21\}}$ at a certified point of $JBOX(R) \times [0, R]$, a covariance eigenvalue below the uniform floor 0.123809288439 on $[0, 1e-5]$, or a conditional moment bound exceeding B_3/B_4 's values, kills the corresponding input and, if load-bearing, the certified pair. The four contract mutations must continue to be rejected (rerun falsify.py: any zero exit).

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